METABOLOMICS EFFECTS OF FOLDING CORRECTION IN RETINITIS PIGMENTOSA RHODOPSIN MUTANT P23A

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PURPOSE

Retinitis pigmentosa (RP) is an inherited retinal degenerative disease characterized by progressive photoreceptor loss, leading to night blindness, tunnel vision, and eventual blindness. RP affects ~1 in 3,000 individuals worldwide and currently has no cure (1). Mutations in the rhodopsin gene, a light-sensitive G protein–coupled receptor (GPCR) essential for phototransduction in rod photoreceptors, are the most common cause of autosomal dominant RP (1). Many mutations disrupt rhodopsin folding, causing endoplasmic reticulum retention, aggregation, and cellular stress. Class II mutations such as P23A and P23H cause misfolding while initially preserving rod viability, making them attractive therapeutic targets. Structural stabilization through the N2C/D282C disulfide bond can restore rhodopsin folding and stability (2) (Figure 1). Emerging evidence suggests that metabolic dysfunction contributes to RP progression (3), but the relationship between rhodopsin misfolding and cellular metabolism remains poorly understood**. We hypothesize that rhodopsin misfolding (P23A) drives metabolic dysregulation, and that structural rescue (P23A/N2C/D282C) restores key metabolic pathways.**

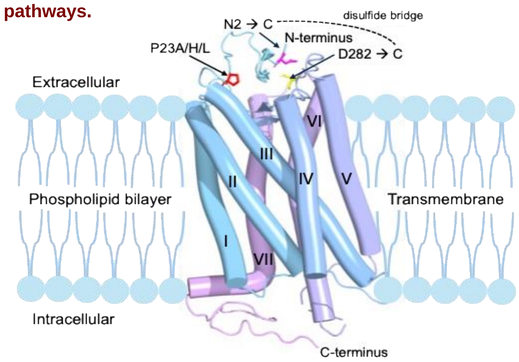


Figure 1. Structural representation of rhodopsin P23A mutation and N2C/D282C disulfide bond correction strategy. N2C and D282C are two additional engineered substitutions introduced on the P23A background (marked in pink and yellow, respectively) form a stabilizing intramolecular disulfide bridge that structurally corrects the misfolding caused by the mutation at position P23.

METHODS

Experimental overview is shown in Figure 2. **Inducible Rhodopsin Cell Model:** Stable HEK293S-GnTI⁻ TetR cell lines expressing Wild type (WT), P23A, or structurally rescued P23A/N2C/D282C rhodopsin were used (4). Expression was induced with tetracycline (2 µg/mL) and sodium butyrate (5 mM) for 24 h. **Cell Culture and Experimental Design:** Cells were cultured in DMEM with 10% FBS and 1% penicillin–streptomycin in 6-well plates and harvested after induction. Empty vector controls and induction controls were included for baseline correction. Overview shown in Figure 3. **Protein Expression Validation:** Rhodopsin expression was verified by SDS-PAGE and Western blot using RHO 1D4 antibody, with GAPDH as a loading control. **Metabolite Extraction and GC–MS Analysis:** Polar metabolites were extracted from ~10^6 cells, derivatized, and analyzed using an Agilent 7820A GC–5977B MSD system. Metabolites were identified using an internal library of 126 metabolites. **Data Processing and Statistical Analysis:** Metabolomic data were processed using MetaboAnalyst 5.0. Data were log-transformed, normalized, and batch-corrected (ComBat). PCA, PLS-DA, VIP scoring, ANOVA, and pathway enrichment analysis were used to identify metabolite and pathway alterations.

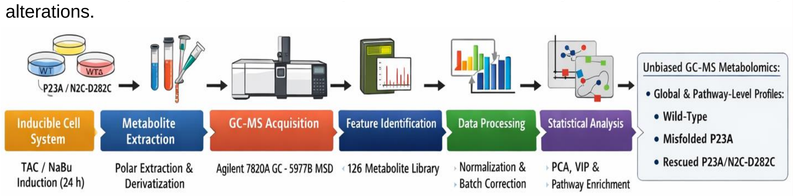
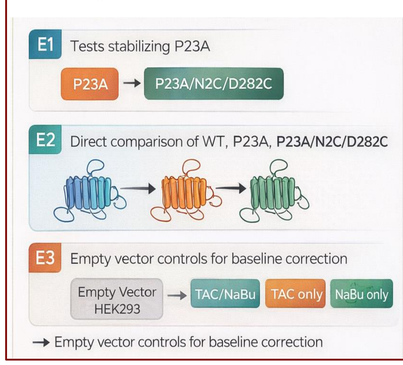


Figure 2. Experimental workflow for untargeted GC–MS metabolomics analysis of rhodopsin misfolding and structural rescue.



Untargeted metabolomics analysis of RP mutant P23A, and its structural correction variant in inducible rhodopsin-expressing cells, isolates folding-driven effects from chromophore binding.

Figure 3. Experimental design for evaluating metabolic effects of rhodopsin misfolding and structural rescue. Inducible, multi-control design isolates metabolic signatures driven specifically by rhodopsin misfolding and its structural correction

RESULTS

Rhodopsin expression: Western blot analysis confirmed robust TAC/NaBu-induced rhodopsin expression across three independent experiments (Figure 4). No rhodopsin band was detected in untreated samples, confirming induction specificity. WT and P23A/N2C/D282C showed diffuse banding patterns at the expected molecular weight, consistent with heterogeneous glycosylation of properly folded rhodopsin. In contrast, the P23A mutant displayed altered band intensities, consistent with protein instability and misfolding. Empty vector HEK293 controls showed no signal, confirming antibody specificity, while GAPDH confirmed equal loading. Inducible rhodopsin expression is reproducible across independent experiments. **P23A misfolding reduces stable rhodopsin expression, while N2C/D282C structural correction restores expression levels comparable to WT.**

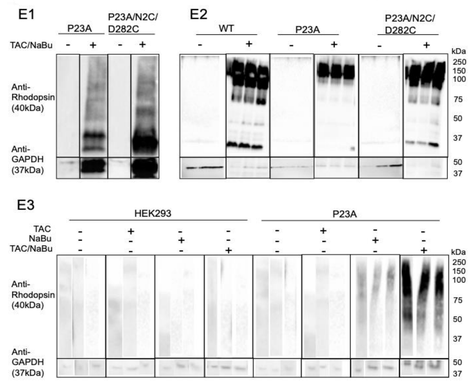


Figure 4. Western blot analysis of rhodopsin using the antibody anti-1D4 and anti-GAPDH expression across different experimental groups. In three experiments, E1, E2 and E3, we compared HEK293 empty vector cells, WT, P23A, and P23A/N2C/D282C stable cell lines. (A) In E1, P23A, and P23A/N2C/D28C cells treated with Tac/NaBu. (B) In E2, WT, P23A, and P23A/N2C/D28C treated with Tac/NaBu. (C) In E3, HEK293 and P23A treated with Tac only, NaBu only, and Tac/NaBu.

Distinct metabolic clustering of WT, P23A, and structurally rescued P23A/N2C/D282C: Principal component analysis (PCA) revealed clear separation of P23A TAC/NaBu samples from WT and P23A/N2C/D282C, indicating a distinct metabolic signature associated with rhodopsin misfolding (Figure 5). In contrast, P23A/N2C/D282C clustered closely with WT, suggesting restoration of a WT-like metabolic profile following structural rescue. HEK293 empty vector controls showed minimal metabolic changes with TAC/NaBu treatment, confirming that the inducer itself does not substantially alter metabolism. PERMANOVA analysis demonstrated a significant genotype effect on global metabolite composition ($R^2 \approx 0.94$, $p = 0.001$), with P23A showing the strongest divergence. **P23A formed a distinct branch while P23A/N2C/D282C grouped with WT, supporting partial metabolic restoration after structural correction.**

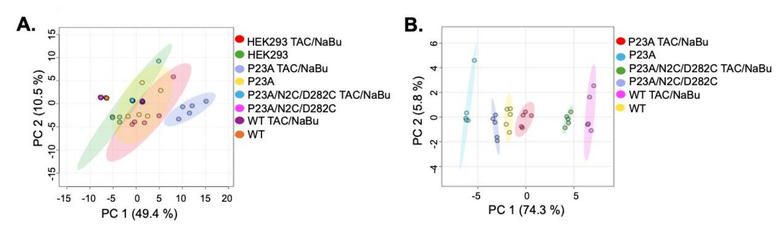


Figure 5. PCA and hierarchical clustering of metabolomic profiles (A) PCA showing partial overlap and group-wise separation among all batch-corrected experimental groups E1-E3. (B) PCA of E 1 and 2 showing P23A Tac/NaBu clustering distinctly from both WT and P23A/N2C/D282C Tac/NaBu.

Differential metabolite signatures in WT, P23A, and P23A/N2C/D282C: Pairwise metabolite analysis of TAC/NaBu-treated WT, P23A, and P23A/N2C/D282C revealed distinct metabolic profiles (Figure 6). P23A showed widespread metabolite downregulation, indicating global metabolic suppression associated with rhodopsin misfolding. In contrast, P23A/N2C/D282C displayed increased metabolite levels relative to P23A, suggesting partial metabolic restoration. **Structural rescue shifted the metabolic profile toward a WT-like state. After removing background and treatment-related metabolites, six metabolites remained uniquely associated with P23A TAC/NaBu cells: adenosine, L-allothreonine, L-cystine, malonic acid, protocatechuic acid, and 4-hydroxyproline.**

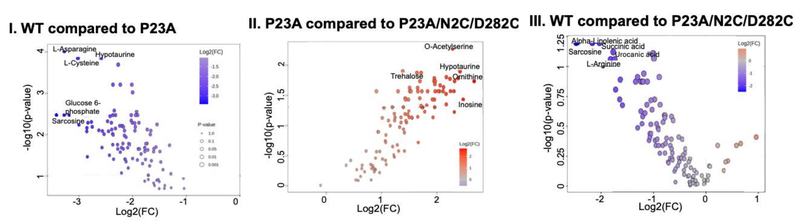


Figure 6. Volcano plot showing significantly upregulated/downregulated metabolites. WT compared to P23A, showed 99 significantly downregulated metabolites in P23A. II. P23A compared to P23A/N2C/D282C, showed 68 significantly upregulated metabolites in P23A/N2C/D282C. III. P23A/N2C/D282C compared to WT, showed 9 downregulated metabolites in P23A/N2C/D282C.

Metabolic suppression in P23A and partial rescue in P23A/N2C/D282C: Heatmap and VIP analyses revealed clear metabolic differences across WT, P23A, and P23A/N2C/D282C (Figure 7). P23A TAC/NaBu cells showed broadly reduced metabolite levels, indicating metabolic suppression associated with rhodopsin misfolding. In contrast, P23A/N2C/D282C exhibited an intermediate profile shifting toward WT, suggesting partial metabolic restoration. **Key discriminating metabolites included sarcosine, L-arginine, succinic acid, and urocanic acid.**

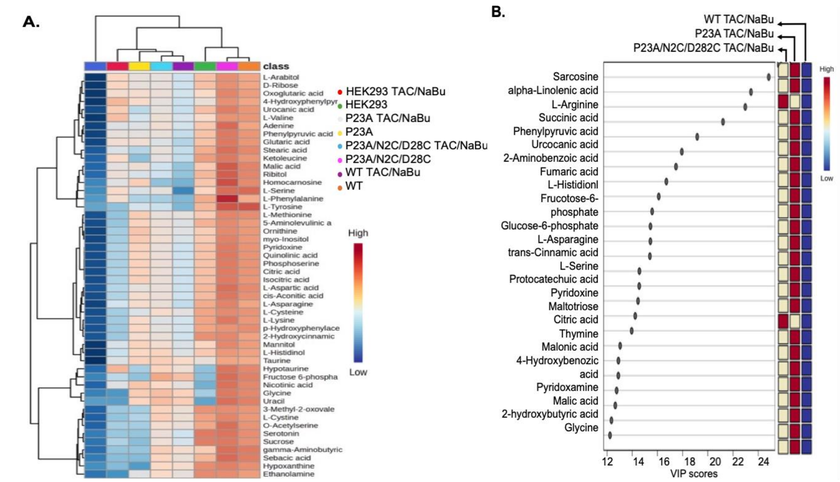


Figure 7. Multivariate analysis reveals distinct metabolic signatures associated with rhodopsin misfolding and structural correction. (A) Heatmap of relative changes in the metabolite abundance across all experiments, highlighting the metabolite expression patterns associated with rhodopsin misfolding and its correction. (B) Variable Importance in Projection (VIP) plot showing the top 25 metabolites that contributed most significantly (VIP score > 1.2) to the group separation based on partial least squares-discriminant analysis (PLS-DA). .

Metabolic pathways disrupted by rhodopsin misfolding: Pathway enrichment analysis showed that P23A rhodopsin misfolding disrupts key metabolic pathways, several of which shift toward recovery in P23A/N2C/D282C (Figure 8). **The most affected pathways included arginine biosynthesis, arginine and proline metabolism, taurine and hypotaurine metabolism, and glutathione metabolism, consistent with redox imbalance and mitochondrial stress associated with misfolded rhodopsin.**

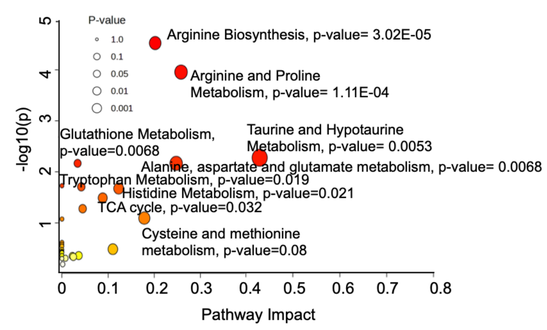


Figure 8. Metabolic pathway enrichment reveals key perturbations in misfolded rhodopsin-expressing cells. Pathway impact analysis using MetaboAnalyst highlights the most significantly dysregulated metabolic pathways in P23A compared to WT and P23A/N2C/D282C.

CONCLUSIONS

- First untargeted GC–MS metabolomics study examining metabolic consequences of rhodopsin misfolding using inducible WT, P23A, and P23A/N2C/D282C cell lines. Structural rescue (P23A/N2C/D282C) partially restores metabolic profiles toward WT, linking protein folding integrity to cellular metabolism.
- Misfolded P23A rhodopsin induces global metabolic suppression, affecting amino acid metabolism, redox balance, and mitochondrial pathways. Key disrupted pathways include arginine/proline metabolism, glutathione metabolism, taurine metabolism, and TCA cycle intermediates.
- Future work: validate key metabolites, perform metabolic flux and mitochondrial assays, and extend studies to retinal models and therapeutic metabolic interventions.

Works cited: [1] Zhen, F. et al. Rhodopsin-associated retinal dystrophy: Disease mechanisms and therapeutic strategies. Front. Neurosci. 17, (2023). [2] Opefi, C. A., South, K., Reynolds, C. A., Smith, S. O. & Reeves, P. J. Retinitis Pigmentosa Mutants Provide Insight into the Role of the N-terminal Cap in Rhodopsin Folding, Structure, and Function. J Biol Chem 288, 33912–33926 (2013). [3] Athanasiou, D. et al. The molecular and cellular basis of rhodopsin retinitis pigmentosa reveals potential strategies for therapy. Prog Retin Eye Res 62, 1–23 (2018). [4] Reeves, P. J., Kim, J.-M. & Khorana, H. G. Structure and function in rhodopsin: A tetracycline-inducible system in stable mammalian cell lines for high-level expression of opsin mutants. Proc Natl Acad Sci U S A 99, 13413–13418 (2002). The work was supported by Arizona State University startup funds and a Graduate Completion Fellowship. Email: Judith.Klein-Seetharaman@asu.edu